

Model Paper - Database Management Systems

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Q1

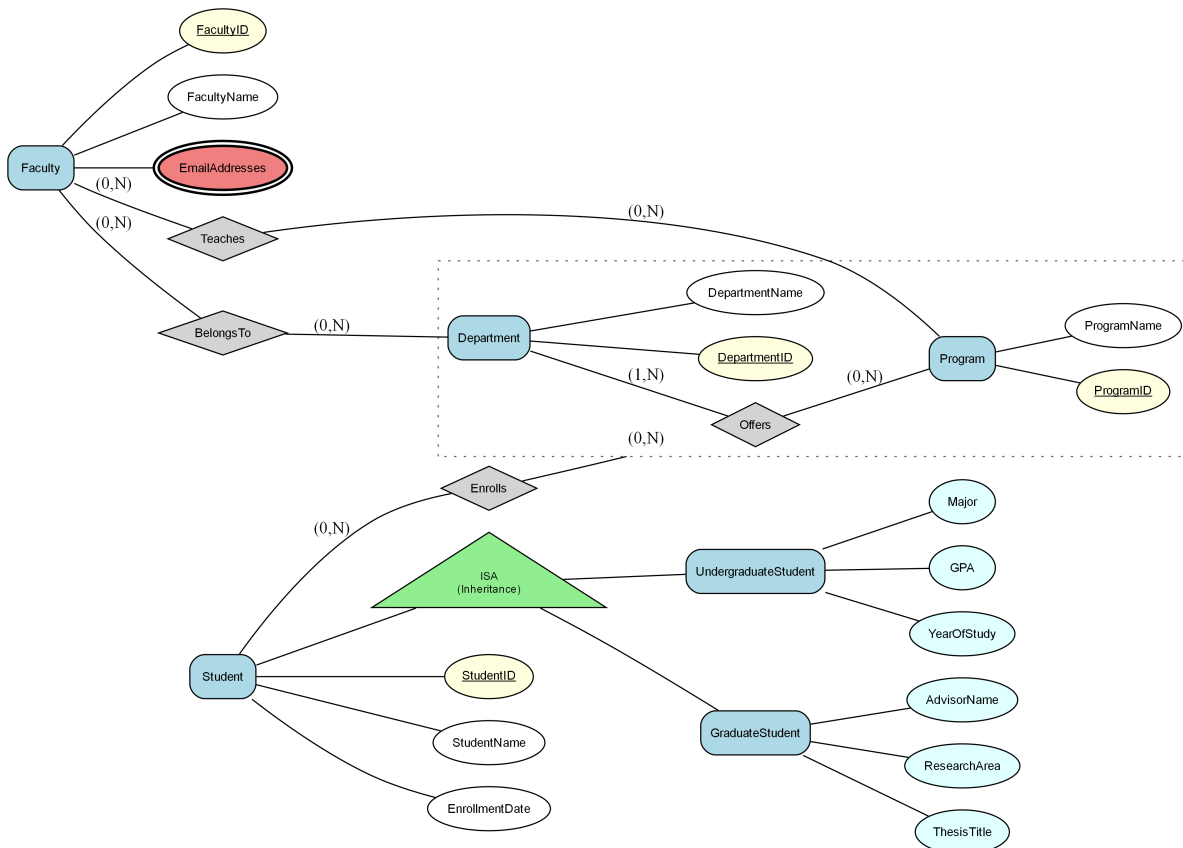
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

Consider a library management system that tracks books, authors, patrons, and loans. The 'Book' entity has attributes: BookID, Title, ISBN, and Genre. The 'Author' entity includes AuthorID, Name, and Nationality. Each book can have multiple authors, resulting in a many-to-many relationship between 'Book' and 'Author'. The 'Patron' entity has PatronID, Name, and MembershipDate. Patrons can check out multiple books, but each loan is associated with only one book and one patron, which forms a one-to-many relationship from 'Patron' to 'Loan'. The 'Loan' entity has attributes: LoanID and LoanDate. Additionally, introduce an ISA hierarchy where the 'Patron' can be classified into 'StudentPatron' with an attribute: SchoolName, and 'AdultPatron' with an attribute: EmploymentStatus

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly.
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer.